Glycogen storage disease occurs when there is a deficiency of glucose 6-phosphatase or a genetically-inherited inability to transport glucose 6-phosphatase to the inner surface of microsomes where it normally participates in glucose metabolism.

Because the condition results in the inability to convert glycogen into glucose, the patient is extremely susceptible to hypoglycemia and usually requires oral feedings every 2-3 hours. Symptoms of this disorder include growth impairment, mental retardation, seizures, osteoporosis due to chronic metabolic acidosis, and enlargement of the kidneys and liver due to the accumulation of glycogen in the organs. Platelet dysfunction with a corresponding bleeding tendency may be present and a tendency toward infection may be present due to impairment of neutrophil activity.

These patients require decreased fasting times, administration of exogenous glucose, monitoring of arterial pH and blood glucose concentrations, and strict asepsis.

*Hines RL, Marschall KE. Stoeltings Anesthesia & Co-Existing Diseases. 6th ed. New York, NY: Churchill-Livingston; 2012: 311.*

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